

Name: _____

45°-45°-90° triangles. Give exact answers.

1. Legs 9. Find the hypotenuse.
2. Hypotenuse 16. Find each leg.
3. Hypotenuse $11\sqrt{2}$. Find each leg.

30°-60°-90° triangles. Give exact answers.

4. Shorter leg 7. Find the hypotenuse and longer leg.
5. Longer leg 15. Find the shorter leg and hypotenuse.
6. Hypotenuse 22. Find both legs.

Special triangles hiding in shapes.

7. A square has side 9. Find its diagonal (exact).
8. An equilateral triangle has side 10. Find its altitude (exact).
9. A baseball diamond is a square with 90-ft sides. How far is the throw from home plate to second base, to the nearest tenth?
10. A 45°-45°-90° triangle and a 30°-60°-90° triangle have the same hypotenuse, 12. Which has the longer SHORTER leg? Show both.